Practical 5

Aim: Execution of Memory Management Commands like free,/proc/meminfo, top,htop, df, du, vmstat, demidecode, sar, pagesize, etc.

Theory:

Explain Memory Management Commands ?

Scenario:

Situation 1:

A Linux server is running very slowly. Users report delays while opening applications.

Questions:

1. Which command will you use to check current memory usage?

   The **free -h** command is used to check current memory usage. It displays total, used, free, shared, buffer/cache, and available memory in a human-readable format (MB/GB), making it easy to quickly understand system memory status.

2. How will you check detailed memory information from the system?

   The **cat/proc/meminfo** command provides detailed memory statistics directly from the Linux kernel. It includes information such as total memory, free memory, buffers, cached memory, swap usage, and more, which helps in deeper analysis.

3. How do you monitor real-time memory usage?

   The **top** or **htop** command is used to monitor real-time memory usage. These tools continuously update system statistics and show memory consumption along with running processes, CPU usage, and load average.

4. Which command helps identify if swapping is happening?

   The **vmstat** command helps identify swapping activity. It shows swap-in (**si**) and swap-out (**so**) values, which indicate whether the system is using swap memory.

5. How can you determine if memory bottleneck is the issue?

   You can use **sar -r** and **vmstat** to determine memory bottlenecks. If free memory is very low, swap usage is high, and **si/so** values are non-zero, it indicates the system is under memory pressure and may have a bottleneck.

Expected Commands:

free -h

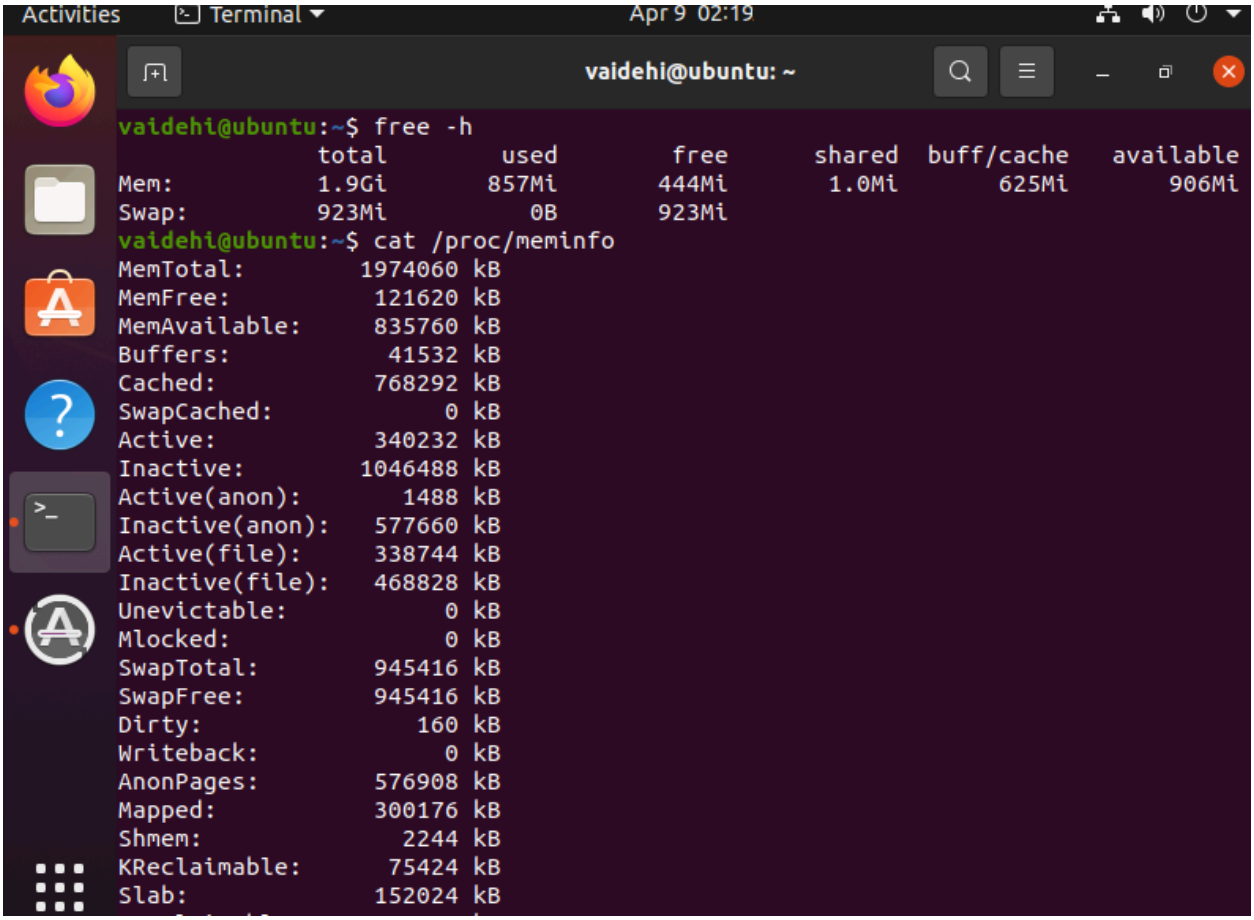
cat /proc/meminfo

top or htop

vmstat

sar -r

Outputs 1:



Activities Terminal Apr 9 02:20

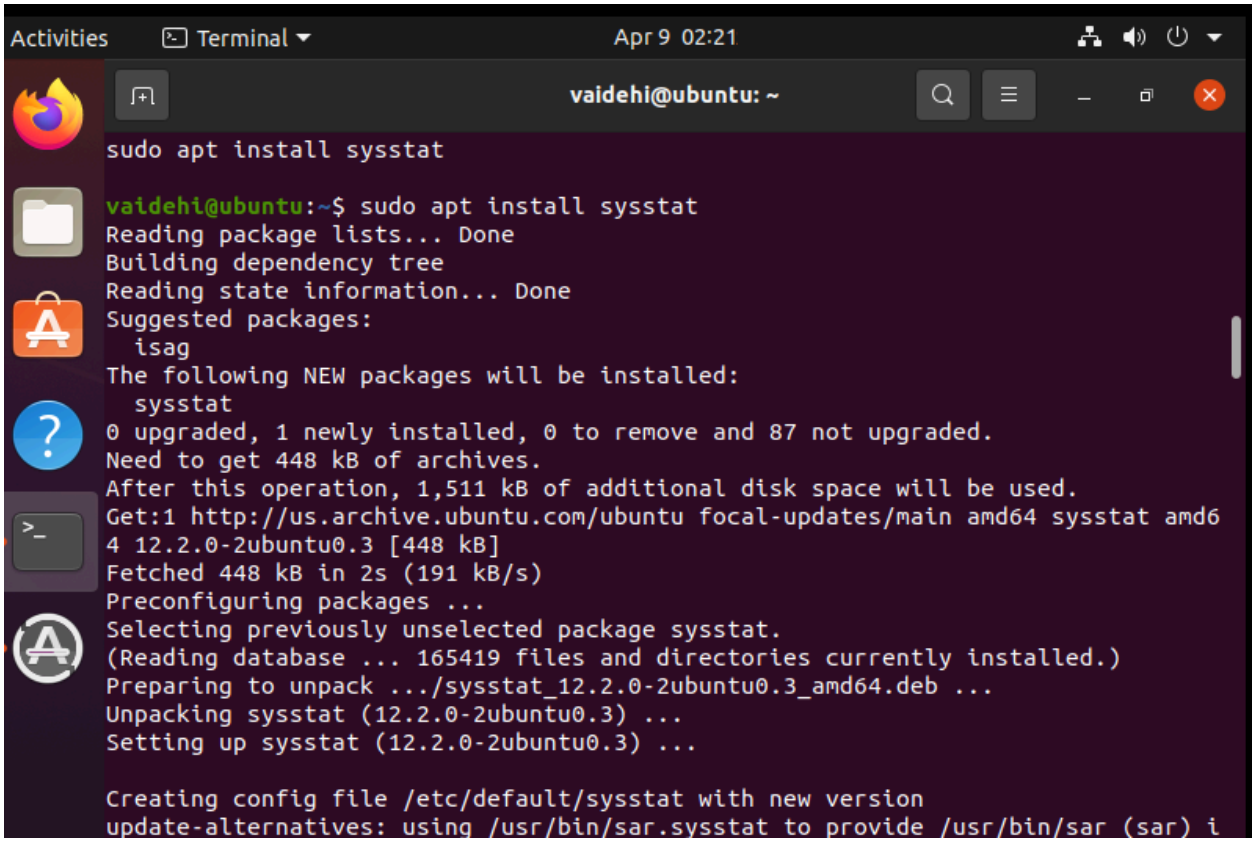
vaidehi@ubuntu: ~

```
DirectMap2M:     1882112 kB
DirectMap1G:           0 kB
vaidehi@ubuntu:~$ top

top - 02:13:05 up 2 min,  1 user,  load average: 0.21, 0.12, 0.04
Tasks: 315 total,   1 running, 314 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.1 us,  3.1 sy,  0.0 ni, 93.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1927.8 total,    118.3 free,    944.7 used,    864.8 buff/cache
MiB Swap:    923.3 total,    923.3 free,      0.0 used.    815.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2199 vaidehi   20   0  289768  68812  41460 S   6.2   3.5   0:01.58 Xorg
      1 root      20   0  102664  11620   8520 S   0.0   0.6   0:02.28 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthrea+
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_pa+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_f+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      7 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworke+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworke+
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.60 kworke+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_per+
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksofti+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.10 rcu_sc+
```

Activities Terminal Apr 9 02:21

vaidehi@ubuntu: ~

```
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migrat+
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.03 ksofti+
vaidehi@ubuntu:~$ htop

Command 'htop' not found, but can be installed with:

sudo snap install htop  # version 3.4.1, or
sudo apt  install htop  # version 2.2.0-2build1

See 'snap info htop' for additional versions.

vaidehi@ubuntu:~$ sudo snap install htop  # version 3.4.1
[sudo] password for vaidehi:
htop 3.4.1 from maxiberta✪ installed
vaidehi@ubuntu:~$ htop
locales-launch: Data of en_US locale not found, generating, please wait...
vaidehi@ubuntu:~$ vmstat
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 1  0   1036 216580  37868 770184    0    2  1311   269  363  750  5  4 90  0  0
vaidehi@ubuntu:~$ sar -r

Command 'sar' not found, but can be installed with:

sudo apt install sysstat
```



Situation 2:

You suspect the system is using too much swap memory.

Questions:

1. How will you check swap usage?

   The **free -h** command shows swap usage clearly, including total, used, and free swap memory.

2. Which field in `/proc/meminfo` shows swap details?

   In `/proc/meminfo`, the fields `SwapTotal` and `SwapFree` show swap memory details. These values help determine how much swap is available and how much is currently being used.

3. How do you monitor swap activity continuously?
   The `vmstat 1` command monitors swap activity continuously by refreshing every second. It shows real-time swap-in (`si`) and swap-out (`so`) values.

4. What does si/so in vmstat indicate?

   In `vmstat`, `si` (swap in) indicates data moved from swap to RAM, and `so` (swap out) indicates data moved from RAM to swap. High values suggest heavy swapping, which can degrade performance.

Outputs 2:

```
vaidehi@ubuntu: ~
vaidehi@ubuntu:~$ sar -r
Cannot open /var/log/sysstat/sa09: No such file or directory
Please check if data collecting is enabled
vaidehi@ubuntu:~$ free -h
              total        used        free      shared  buff/cache   av
ailable
Mem:          1.9Gi       955Mi       122Mi       2.0Mi       849Mi
  791Mi
Swap:         923Mi       1.0Mi       921Mi
vaidehi@ubuntu:~$ cat /proc/meminfo
MemTotal:        1974060 kB
MemFree:          125028 kB
MemAvailable:     810956 kB
Buffers:           67792 kB
Cached:           700516 kB
SwapCached:           92 kB
Active:           459728 kB
Inactive:         886492 kB
Active(anon):       1940 kB
Inactive(anon):   578216 kB
Active(file):     457788 kB
Inactive(file):   308276 kB
Unevictable:           0 kB
Mlocked:               0 kB
SwapTotal:        945416 kB
SwapFree:         943612 kB
Dirty:               308 kB
```

```
vaidehi@ubuntu: ~
Hugetlb:               0 kB
DirectMap4k:      255872 kB
DirectMap2M:     1841152 kB
DirectMap1G:           0 kB
vaidehi@ubuntu:~$ vmstat 2
procs -----------memory---------- ---swap-- -----io---- -system-- ------
cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy
id wa st
 1  0   1804 125028  67792 802532    0    2  1248   347  344  707  5  4
90  1  0
 0  0   1804 125028  67792 802532    0    0     0     0  320  814  2  1
98  0  0
 0  0   1804 125028  67792 802532    0    0     0     0  248  448  1  1
98  0  0
 0  0   1804 125028  67792 802532    0    0     0     0  127  226  1  0
99  0  0
 0  0   1804 124780  67796 802556    0    0    14     0  294  673  2  1
97  0  0
 0  0   1804 124780  67796 802556    0    0     0     0   85  174  0  0
100  0  0
 0  0   1804 124780  67796 802556    0    0     0     0   75  151  1  0
```

Situation 3:

One application is consuming too much memory.

Questions:

1. Which command shows processes sorted by memory usage?

   The `ps aux --sort=-%mem` command lists all processes sorted by highest memory usage at the top, making it easy to identify memory-heavy applications.

2. How can you interactively manage processes?

   The `top` or `htop` command allows interactive process management. You can kill processes, change priority, and monitor memory usage dynamically.

3. How to identify the PID of highest memory consumer?

   Using `ps aux --sort=-%mem`, the process at the top has the highest memory usage. Its PID is shown in the PID column, which can be used to manage or terminate the process.

Expected Commands:

top

htop

ps aux --sort=-%mem

Outputs 3:

```
vaidehi@ubuntu:~$ top

top - 02:18:44 up 8 min,  1 user,  load average: 0.06, 0.12, 0.06
Tasks: 280 total,   1 running, 279 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.6 us,  0.3 sy,  0.0 ni, 98.1 id,  0.0 wa,  0.0 hi,  0.0 si,
 0.0 st
MiB Mem :   1927.8 total,    112.1 free,    965.4 used,    850.3 buff/ca
che
MiB Swap:    923.3 total,    921.5 free,      1.8 used.    782.3 avail M
em

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ C
OMMAND
   2199 vaidehi   20   0  289768  68572  41220 S   1.9   3.5   0:05.18 X
org
   2327 vaidehi   20   0 3811788 249112 102124 S   1.3  12.6   0:06.40 g
nome-+
      1 root      20   0  169552  12980   8520 S   0.0   0.7   0:02.99 s
ystemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 k
threa+
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 r
cu_gp
```

```
vaidehi@ubuntu: ~
0  0   1804 124780  67832 802568    0    0     0     0  317  599  2  2
S  0  0
C
aidehi@ubuntu:~$ ps aux --sort=-%mem
SER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMA
D
aidehi     2327  1.5 12.6 3811788 249076 ?      Ssl  02:11   0:06 /usr/
in/gno
aidehi     2804  1.1  9.5 956240 188340 ?       SNl  02:12   0:03 /usr/
in/pyt
aidehi     2199  1.1  3.4 289768 68572 tty2     Sl+  02:11   0:04 /usr/
ib/xor
aidehi     2515  0.0  2.9 645848 58472 ?        Sl   02:11   0:00 /usr/
ibexec
aidehi     2637  0.4  2.5 815996 50136 ?        Ssl  02:11   0:01 /usr/
```

```
vaidehi@ubuntu: ~
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.11 k
softi+
vaidehi@ubuntu:~$ htop
vaidehi@ubuntu:~$ ps aux --sort=-%mem
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMA
ND
vaidehi     2327  1.4 12.6 3811788 249112 ?      Ssl  02:11   0:06 /usr/
bin/gno
vaidehi     2804  0.9  9.5 956240 188340 ?       SNl  02:12   0:03 /usr/
bin/pyt
vaidehi     2199  1.1  3.4 289768 68572 tty2     Sl+  02:11   0:05 /usr/
lib/xor
vaidehi     2515  0.0  2.9 645848 58472 ?        Sl   02:11   0:00 /usr/
libexec
vaidehi     2637  0.4  2.5 815996 50808 ?        Rsl  02:11   0:02 /usr/
libexec
root         801  0.7  2.1 1850300 42256 ?       Ssl  02:10   0:03 /usr/
lib/sna
vaidehi     2489  0.2  2.0 143712 40648 ?        Sl   02:11   0:01 /usr/
bin/vmt
vaidehi     2162  0.0  1.7 553264 34284 ?        Sl   02:11   0:00 /usr/
libexec
vaidehi     2351  0.1  1.6 275692 31632 ?        Sl   02:11   0:00 /usr/
libexec
vaidehi     2664  0.0  1.5 494092 31280 ?        Sl   02:12   0:00 updat
```

Conclusion:

This practical demonstrates how Linux memory management commands help monitor system performance, detect memory issues, and identify resource-heavy processes. Tools like `free`, `top`, `vmstat`, and `sar` are essential for diagnosing memory bottlenecks and ensuring efficient system operation.